SQL JOINS PRACTICE WORKBOOK

Questions with SQL Queries and Output

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Topic:	**SQL Joins (MySQL)**
Software Used:	**MySQL Workbench 8.0**
Date:	**23 July 2026**

Introduction

This document contains SQL JOIN practice questions along with their SQL queries and corresponding output screenshots executed in MySQL Workbench. The database, questions, and expected outputs are based on the provided SQL Joins Practice Workbook.

Database Creation and Table Setup

Create Tables

```
CREATE TABLE departments (
  dept_id INT PRIMARY KEY,
  dept_name VARCHAR(50),
  location VARCHAR(50)
);

CREATE TABLE employees (
  emp_id INT PRIMARY KEY,
  emp_name VARCHAR(50),
  dept_id INT,
  salary INT,
  city VARCHAR(50)
);

CREATE TABLE projects (
  proj_id INT PRIMARY KEY,
  proj_name VARCHAR(50),
  dept_id INT,
  budget INT
);
```

Insert Data

```
INSERT INTO departments VALUES
(10,'HR','Jaipur'),
(20,'IT','Bangalore'),
```

```
(30,'Finance','Mumbai'),
(40,'Marketing','Delhi'),
(50,'Legal','Pune');

INSERT INTO employees VALUES
(101,'Amit',10,45000,'Jaipur'),
(102,'Bhavna',20,72000,'Bangalore'),
(103,'Chirag',20,68000,'Bangalore'),
(104,'Divya',30,55000,'Mumbai'),
(105,'Esha',NULL,39000,'Delhi'),
(106,'Farhan',40,61000,'Delhi'),
(107,'Gaurav',60,50000,'Kolkata'),
(108,'Hina',30,83000,'Mumbai');

INSERT INTO projects VALUES
(1,'Payroll Revamp',10,120000),
(2,'Cloud Migration',20,500000),
(3,'Mobile App',20,300000),
(4,'Audit Automation',30,250000),
(5,'Brand Refresh',40,180000),
(6,'Data Lake',70,400000);
```

Question No. 1

List each employee with their department name and location.

```sql
-- 1

SELECT e.emp_id, e.emp_name, d.dept_name, d.location
FROM employees e
INNER JOIN departments d
ON e.dept_id = d.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_id	emp_name	dept_name	location
101	Amit	HR	Jaipur
102	Bhavna	IT	Bangalore
103	Chirag	IT	Bangalore
104	Divya	Finance	Mumbai
106	Farhan	Marketing	Delhi
108	Hina	Finance	Mumbai

Question No. 2

Show every project along with the name of the department that owns it.

```sql
-- 2

SELECT p.proj_id, p.proj_name, d.dept_name, p.budget
FROM projects p
INNER JOIN departments d
ON p.dept_id = d.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

proj_id	proj_name	dept_name	budget
1	Payroll Revamp	HR	120000
2	Cloud Migration	IT	500000
3	Mobile App	IT	300000
4	Audit Automation	Finance	250000
5	Brand Refresh	Marketing	180000

Question No. 3

Show employee names together with the projects run by their own department.

```sql
-- 3

SELECT e.emp_name, d.dept_name, p.proj_name
FROM employees e
INNER JOIN departments d
ON e.dept_id = d.dept_id
INNER JOIN projects p
ON d.dept_id = p.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	dept_name	proj_name
Amit	HR	Payroll Revamp
Bhavna	IT	Mobile App
Bhavna	IT	Cloud Migration
Chirag	IT	Mobile App
Chirag	IT	Cloud Migration
Divya	Finance	Audit Automation
Farhan	Marketing	Brand Refresh
Hina	Finance	Audit Automation

Question No. 4

List ALL employees with their department name; show NULL when the employee has no matching department.

```sql
-- 4

SELECT e.emp_id, e.emp_name, e.dept_id, d.dept_name
FROM employees e
LEFT JOIN departments d
ON e.dept_id = d.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_id	emp_name	dept_id	dept_name
101	Amit	10	HR
102	Bhavna	20	IT
103	Chirag	20	IT
104	Divya	30	Finance
105	Esha	NULL	NULL
106	Farhan	40	Marketing
107	Gaurav	60	NULL
108	Hina	30	Finance

Question No. 5

List ALL departments with their projects; departments with no project must still appear.

```sql
-- 5

SELECT d.dept_id, d.dept_name, p.proj_name, p.budget
FROM departments d
LEFT JOIN projects p
ON d.dept_id = p.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_id	dept_name	proj_name	budget
10	HR	Payroll Revamp	120000
20	IT	Mobile App	300000
20	IT	Cloud Migration	500000
30	Finance	Audit Automation	250000
40	Marketing	Brand Refresh	180000
50	Legal	NULL	NULL

Question No. 6

Find only those employees who do NOT belong to any valid department.

```sql
-- 6

SELECT e.emp_id, e.emp_name, e.dept_id
FROM employees e
LEFT JOIN departments d
ON e.dept_id = d.dept_id
WHERE d.dept_id IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_id	emp_name	dept_id
105	Esha	NULL
107	Gaurav	60

Question No. 7

List ALL departments and any employees in them, using a RIGHT JOIN with employees on the left.

```sql
-- 7

SELECT e.emp_name, d.dept_id, d.dept_name
FROM employees e
RIGHT JOIN departments d
ON e.dept_id = d.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	dept_id	dept_name
Amit	10	HR
Chirag	20	IT
Bhavna	20	IT
Hina	30	Finance
Divya	30	Finance
Farhan	40	Marketing
NULL	50	Legal

Question No. 8

List ALL projects and the department that owns them, keeping projects that point to a missing department.

```sql
-- 8

SELECT d.dept_name, p.proj_id, p.proj_name, p.dept_id
FROM departments d
RIGHT JOIN projects p
ON d.dept_id = p.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_name	proj_id	proj_name	dept_id
HR	1	Payroll Revamp	10
IT	2	Cloud Migration	20
IT	3	Mobile App	20
Finance	4	Audit Automation	30
Marketing	5	Brand Refresh	40
NULL	6	Data Lake	70

Question No. 9

Find departments that currently have NO employee assigned.

```sql
-- 9

SELECT d.dept_id, d.dept_name, d.location
FROM departments d
LEFT JOIN employees e
ON d.dept_id = e.dept_id
WHERE e.emp_id IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_id	dept_name	location
50	Legal	Pune

Question No. 10

Produce every possible pairing of the Finance/Legal departments with employees earning above 70000.

```sql
-- 10

SELECT e.emp_name, e.salary, d.dept_name
FROM employees e
CROSS JOIN departments d
WHERE e.salary > 70000
AND d.dept_name IN ('Finance', 'Legal');
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	salary	dept_name
Bhavna	72000	Legal
Bhavna	72000	Finance
Hina	83000	Legal
Hina	83000	Finance

Question No. 11

Pair every project having a budget of at least 400000 with every department located in Mumbai or Pune.

```sql
-- 11

SELECT p.proj_name, p.budget, d.dept_name, d.location
FROM projects p
CROSS JOIN departments d
WHERE p.budget >= 400000
AND d.location IN ('Mumbai','Pune');

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

proj_name	budget	dept_name	location
Data Lake	400000	Finance	Mumbai
Cloud Migration	500000	Finance	Mumbai
Data Lake	400000	Legal	Pune
Cloud Migration	500000	Legal	Pune

Question No. 12

How many total row combinations result from CROSS JOIN of employees and departments? Show the count.

```sql
-- 12

SELECT COUNT(*) AS total_combinations
FROM employees
CROSS JOIN departments;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_combinations
40

Question No. 13

Show employees whose salary is greater than 60000.

```sql
-- 13

SELECT *
FROM employees
WHERE salary > 60000;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	dept_id	salary	city
102	Bhavna	20	72000	Bangalore
103	Chirag	20	68000	Bangalore
106	Farhan	40	61000	Delhi
108	Hina	30	83000	Mumbai
NULL	NULL	NULL	NULL	NULL

Question No. 14

Show all employees based in Delhi or Mumbai.

```sql
-- 14

SELECT emp_id, emp_name, salary, city
FROM employees
WHERE city IN ('Delhi','Mumbai');
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	salary	city
104	Divya	55000	Mumbai
105	Esha	39000	Delhi
106	Farhan	61000	Delhi
108	Hina	83000	Mumbai
NULL	NULL	NULL	NULL

Question No. 15

Show employees whose dept_id is NULL (unassigned employees).

```sql
-- 15

SELECT emp_id, emp_name, dept_id, salary
FROM employees
WHERE dept_id IS NULL;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	dept_id	salary
105	Esha	NULL	39000
NULL	NULL	NULL	NULL

Question No. 16

Show departments located in Bangalore or Delhi.

```sql
-- 16

SELECT *
FROM departments
WHERE location IN ('Bangalore','Delhi');
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

dept_id	dept_name	location
20	IT	Bangalore
40	Marketing	Delhi
NULL	NULL	NULL

Question No. 17

Show departments whose dept_id is 30 or higher.

```
-- 17

SELECT *
FROM departments
WHERE dept_id >= 30;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

dept_id	dept_name	location
30	Finance	Mumbai
40	Marketing	Delhi
50	Legal	Pune
NULL	NULL	NULL

Question No. 18

Show departments whose name starts with the letter "F" or "L".

```
-- 18

SELECT *
FROM departments
WHERE dept_name LIKE 'F%'
OR dept_name LIKE 'L%';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

dept_id	dept_name	location
30	Finance	Mumbai
50	Legal	Pune
NULL	NULL	NULL

Question No. 19

Show projects with a budget between 200000 and 400000 (inclusive).

```sql
-- 19

SELECT *
FROM projects
WHERE budget BETWEEN 200000 AND 400000;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

proj_id	proj_name	dept_id	budget
3	Mobile App	20	300000
4	Audit Automation	30	250000
6	Data Lake	70	400000
NULL	NULL	NULL	NULL

Question No. 20

Show all projects that belong to dept_id 20.

```sql
-- 20

SELECT *
FROM projects
WHERE dept_id = 20;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

proj_id	proj_name	dept_id	budget
2	Cloud Migration	20	500000
3	Mobile App	20	300000
NULL	NULL	NULL	NULL

Question No. 21

Show projects whose name contains the word "a" and budget is under 300000.

```sql
-- 21

SELECT proj_id, proj_name, budget
FROM projects
WHERE proj_name LIKE '%a%'
AND budget < 300000;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

proj_id	proj_name	budget
1	Payroll Revamp	120000
4	Audit Automation	250000
5	Brand Refresh	180000
NULL	NULL	NULL

Question No. 22

Show employee name, department name, location and project name for all matching rows across all three tables.

```sql
-- 22

SELECT e.emp_name, d.dept_name, d.location, p.proj_name, p.budget
FROM employees e
INNER JOIN departments d
ON e.dept_id = d.dept_id
INNER JOIN projects p
ON d.dept_id = p.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	dept_name	location	proj_name	budget
Amit	HR	Jaipur	Payroll Revamp	120000
Bhavna	IT	Bangalore	Mobile App	300000
Bhavna	IT	Bangalore	Cloud Migration	500000
Chirag	IT	Bangalore	Mobile App	300000
Chirag	IT	Bangalore	Cloud Migration	500000
Divya	Finance	Mumbai	Audit Automation	250000
Farhan	Marketing	Delhi	Brand Refresh	180000
Hina	Finance	Mumbai	Audit Automation	250000

Question No. 23

Show ALL employees, plus department and project details where available (keep employees even with no dept/project).

```sql
-- 23

SELECT e.emp_id, e.emp_name, d.dept_name, p.proj_name
FROM employees e
LEFT JOIN departments d
ON e.dept_id = d.dept_id
LEFT JOIN projects p
ON d.dept_id = p.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_id	emp_name	dept_name	proj_name
101	Amit	HR	Payroll Revamp
102	Bhavna	IT	Mobile App
102	Bhavna	IT	Cloud Migration
103	Chirag	IT	Mobile App
103	Chirag	IT	Cloud Migration
104	Divya	Finance	Audit Automation
105	Esha	NULL	NULL
106	Farhan	Marketing	Brand Refresh
107	Gaurav	NULL	NULL
108	Hina	Finance	Audit Automation

Question No. 24

Show employees earning more than 60000 along with department and any project over 250000 budget.

```sql
-- 24

SELECT e.emp_name, e.salary, d.dept_name, p.proj_name, p.budget
FROM employees e
INNER JOIN departments d
ON e.dept_id = d.dept_id
INNER JOIN projects p
ON d.dept_id = p.dept_id
WHERE e.salary > 60000
AND p.budget > 250000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	salary	dept_name	proj_name	budget
Bhavna	72000	IT	Mobile App	300000
Bhavna	72000	IT	Cloud Migration	500000
Chirag	68000	IT	Mobile App	300000
Chirag	68000	IT	Cloud Migration	500000

Question No. 25

List ALL departments with their employees and projects, including departments having neither.

```
-- 25

SELECT d.dept_id, d.dept_name, e.emp_name, p.proj_name
FROM departments d
LEFT JOIN employees e
ON d.dept_id = e.dept_id
LEFT JOIN projects p
ON d.dept_id = p.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_id	dept_name	emp_name	proj_name
10	HR	Amit	Payroll Revamp
20	IT	Chirag	Mobile App
20	IT	Chirag	Cloud Migration
20	IT	Bhavna	Mobile App
20	IT	Bhavna	Cloud Migration
30	Finance	Hina	Audit Automation
30	Finance	Divya	Audit Automation
40	Marketing	Farhan	Brand Refresh
50	Legal	NULL	NULL

Question No. 26

Show employees who work in a department located in Bangalore or Mumbai, along with the projects of that department.

```
-- 26

SELECT e.emp_name, d.location, p.proj_name
FROM employees e
INNER JOIN departments d
ON e.dept_id = d.dept_id
INNER JOIN projects p
ON d.dept_id = p.dept_id
WHERE d.location IN ('Bangalore','Mumbai');
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	location	proj_name
Bhavna	Bangalore	Mobile App
Bhavna	Bangalore	Cloud Migration
Chirag	Bangalore	Mobile App
Chirag	Bangalore	Cloud Migration
Divya	Mumbai	Audit Automation
Hina	Mumbai	Audit Automation

Question No. 27

Pair employees who work in the same city (avoid duplicate pairs and self-pairing).

```sql
-- 27

SELECT e1.emp_name AS employee_1,
       e2.emp_name AS employee_2,
       e1.city
FROM employees e1
INNER JOIN employees e2
ON e1.city = e2.city
AND e1.emp_id < e2.emp_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee_1	employee_2	city
Bhavna	Chirag	Bangalore
Divya	Hina	Mumbai
Esha	Farhan	Delhi

Question No. 28

Show every project along with employees of that department; keep projects with no employees.

```sql
-- 28

SELECT p.proj_name, p.dept_id, e.emp_name
FROM projects p
LEFT JOIN employees e
ON p.dept_id = e.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

proj_name	dept_id	emp_name
Payroll Revamp	10	Amit
Cloud Migration	20	Chirag
Cloud Migration	20	Bhavna
Mobile App	20	Chirag
Mobile App	20	Bhavna
Audit Automation	30	Hina
Audit Automation	30	Divya
Brand Refresh	40	Farhan
Data Lake	70	NULL